



أسبوع DUBAI
دبي للذكاء AI
الاصطناعي WEEK

DUBAI AI WEEK HACKATHON: AGENTIC AI

هاكاثون أسبوع دبي للذكاء الاصطناعي

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What is AGENTIC AI?

And how is it different from an AI AGENT?

AGENTIC AI vs AI AGENTS

Not all AI tools operate in the same way. Understanding the difference reshapes how we approach automation and changes our ways of working.

AI Agents perform specific tasks based on user inputs and predefined objectives. They automate workflows but do not generalise beyond their designed functions. Like a virtual assistant that schedules meetings based on explicit instructions. With AI Agents, humans stay in the loop of every action, giving step-by-step directions.

With Agentic AI, humans shift into an "agent coordinator" role overseeing goals rather than micromanaging specific tasks. This type of AI models problems dynamically, using probabilistic reasoning, optimisation, and feedback loops to achieve objectives with minimal intervention. It processes data, adjusts strategies, and iterates based on outcomes. For example, a system that autonomously optimises medication distribution by analysing demand, supply chain logistics, and regional health data, adapting its strategy in real time to improve efficiency.

Businesses that harness Agentic AI will unlock automation beyond repetitive tasks; optimising decision-making and transforming ways of working.

Comparison

AGENTIC AI

Overview:

Go beyond simple task automation. Exhibit a higher degree of autonomy - perceiving, reasoning, planning, acting, and continuously learning with minimal human intervention. Solve complex, dynamic problems by setting and pursuing long-term goals.

Examples:

- Self-driving vehicles
- Supply chain management
- AI workforce

AI AGENTS

Overview:

Perform specific tasks autonomously within defined rules or parameters. Typically reactive systems responding to user inputs and environmental triggers following fixed or rule-based logic.

Examples:

- Customer support chatbot
- Email management agent
- Code suggestion tools

Agentic AI Key Characteristics

There are four key characteristics of an Agentic AI system that differentiate it from other AI systems outlined below:

1. PERCEPTION

Active data gathering of real-time data from the surroundings such as sensors, databases, digital interfaces or other external sources. Constant intake of information allows the system to build an understanding of its environment.

2. REASONING

Utilisation of an advanced reasoning engine (often LLMs) to process and analyse data they acquire. Enables interpretation of complex situations to coordinate across different tools and generate a solution.

3. ACTION

Ability to decide and execute actions. Integrate with external tools and software via APIs to carry out multi-step, goal-driven tasks. Guardrails needed to ensure safe and correct task execution and reduce human intervention.

4. LEARNING

A continuous learning loop allows these systems to improve over time. Through receiving feedback on actions and outcomes, they adjust strategies and adapt over time. The iterative process of learning increases their efficiency and effectiveness in managing complex workflows.

FURTHER READING

Article on the difference between AI agents and agentic AI:

<https://medium.com/@elisowski/ai-agents-vs-agentic-ai-whats-the-difference-and-why-does-it-matter-03159ee8c2b4>

Article on Agentic AI from Nvidia's perspective: <https://blogs.nvidia.com/blog/what-is-agentic-ai/>

Harvard business on Agentic AI: <https://hbr.org/2024/12/what-is-agentic-ai-and-how-will-it-change-work>

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TRAVEL

TRAVEL - HACKER BRIEF

MASTER QUESTION	Rigid travel itineraries are fixed and don't adapt to real time changes in personal preferences location offering climate etc. How do we make travel itineraries adapt in real time to traveler preferences and any changes to the context?
WHAT ARE WE OPTIMISING THE SYSTEM FOR?	Provide a personalised, flexible travel experience. Improve traveller satisfaction by anticipating needs and adjusting plans dynamically. Reduce stress by ensuring activities fit with weather conditions and preferences. Help travellers experience the best of Dubai without the hassle of constant decision-making.
WHAT ARE THE KEY USE CASES?	"I am a traveller in Dubai, looking for a personalised and dynamic travel experience. I need a system that adjusts travel bookings and activities throughout the day / trip based on real-time conditions and preferences."
SOURCE DATA	DX Open Data, Synthetic Data
ACCESS	TBD
PROCESSING	Agentic AI-powered travel concierge that: Adjusts bookings and itineraries in real-time based on preferences and environmental conditions (e.g., temperature, activity level). Real-time Adaptation: AI monitors weather and preferences throughout the day. Seamless Adjustments: Automatically changes bookings, activities, and locations without user intervention. Dubai-Specific Customization: Adapts plans for the unique climate and attractions of Dubai.
TOUCH POINTS / INTERFACES	Web, mobile, smart wearables.
STARTERS FOR TEN	Hot weather in Dubai triggers a shift from an outdoor restaurant to an indoor one, or adjusts the timing of swimming activities to cooler hours. Changes the schedule based on user preferences, such as swapping a sports activity with a more relaxed experience.
STRETCH GOAL	Add family or co-traveller preferences
CONSIDERATIONS	Automatically makes new bookings if needed and communicates changes in real-time to the traveler, with a yes to accept push notification.

TRAVEL - DATA SOURCES

We encourage teams to spend at least 1 day compiling data their Agentic AI systems will need. Whether this be pulling from open datasets or creating synthetic data to work on all routes are accepted and encouraged so long as they are in the public domain. Here is some specific guidance for each project:

Source Type	Details
Mapping APIs	Google maps , OpenStreetMap
Weather APIs	Met office
Accommodation APIs	Booking.com API
Traffic info APIs	TomTom traffic API
Events APIs	Eventbrite
Dubai APIs	Dubai API
POI/Attraction info APIs	TripAdvisor
Ride sharing APIs	Uber
Dining APIs	Opentable API
Local news APIs	UAE news API

We have compiled a list of suggestions for where to get relevant real time data. This list is by no means exhaustive and we highly encourage teams to find their own data sources. If cost or general access to these APIs is not possible we would encourage teams to synthesize some data and package it in their own API to simulate an integration with real time data. Please don't put API keys in your source code on GitHub.

TRAVEL - PERSONAS

We have created 3 personas per challenge to give you some inspiration and guidance as to the types of challenges you will be solving. We would like teams to generate some more personas for the challenge and demonstrate how your tool can solve problems and add value to your users.

Persona 1: Tom & Priya, 36 — British Couple on Luxury Vacation

- **Profile:** Visiting Dubai for 7 days, interested in culture, food, and light adventure.
- **Needs:** A smooth experience with timely adjustments due to weather or energy levels.
- **Challenge:** Frustrated by rigid itineraries that don't respond to heat or personal mood.

Persona 2: Sara, 29 — Influencer from Riyadh

- **Profile:** Wants to create exciting, unique content while in Dubai.
- **Needs:** A smart concierge that finds photogenic, trending spots, and adjusts for ideal times and crowds.
- **Challenge:** Wants experiences that feel custom but without the planning hassle.

Persona 3: David, 52 — C-Level Exec Attending GITEX

- **Profile:** In Dubai for a conference, looking to mix business and downtime effectively.
- **Needs:** A system that adapts free time for sports or dining based on changing work schedules.
- **Challenge:** Needs a high-end, frictionless service that respects both spontaneity and professionalism.

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FINANCE

FINANCE- HACKER BRIEF

MASTER QUESTION	TRADITIONAL SAVINGS PLANS DON'T ADAPT TO FLUCTUATING EXPENSES LIKE CHILDCARE, TRAVEL, OR ENTERTAINMENT. How can agentic AI detect real time spending patterns and create automated allocations to adjust savings without manual intervention?
WHAT ARE WE OPTIMISING THE SYSTEM FOR?	Reduce financial stress by optimising savings dynamically. Encourage responsible spending while still enabling enjoyment. Help users consistently meet mortgage goals without lifestyle sacrifices. Provide clear financial insights, empowering better decision-making.
WHAT ARE THE KEY USE CASES?	"I am a young professional with a fluctuating income and changing expenses and I want to buy a property or manage my mortgage payments. I need a system that dynamically adjusts savings contributions while allowing room for discretionary spending."
SOURCE DATA	DX Open Data, Synthetic Data
ACCESS	TBD
PROCESSING	Agentic AI-powered financial concierge that: Analyses spending patterns and adjusts savings accordingly. Creates two funds: 'Play Pot' (fun spending) & 'Save Pot' (mortgage savings). Automatically allocates funds based on life events like childcare expenses or vacation planning. Provides monthly explanations on savings adjustments, ensuring transparency and trust.
TOUCH POINTS / INTERFACES	Web, mobile, smart wearables.
STARTERS FOR TEN	Real-time Adaptability: AI detects spending patterns and life changes. Automated Allocations: Adjusts savings without manual intervention. User Guidance & Transparency: Provides insights on why changes occur.
STRETCH GOAL	Add partner or friends' pots.
CONSIDERATIONS	Automatically allocate to savings pots without intervention.

FINANCE- DATA SOURCES

We encourage teams to spend at least 1 day compiling data their Agentic AI systems will need. Whether this be pulling from open datasets or creating synthetic data to work on all routes are accepted and encouraged so long as they are in the public domain. Here is some specific guidance for each project:

For the finance challenge using real financial data is not possible so teams must synthesize their own data or find open datasets. We have compiled a few datasets for guidance and a paper outlining some techniques for creating synthetic financial data.

Datasets

Synthetic financial data on GitHub: [link](#)

Teams encouraged to explore [kaggle](#) for open datasets

Guidance

Article on generating synthetic financial data: [link](#)

Live Data Sources

Here is a list of potential live data sources teams can use. This list is by no means exhaustive and if access to any is restricted teams are encouraged to synthesize some data and create a mock API. Please don't put API keys in your source code on GitHub.

Source Type	Details
Interest rates API	Trading economics API
Stock market API	Alpha vantage API , Polygon IO
Live news API	Bing news API , Reuters API
Property market API	EstateAgent API
Crypto API	CoinMarketCap API

FINANCE- PERSONAS

We have created 3 personas per challenge to give you some inspiration and guidance as to the types of challenges you will be solving. We would like teams to generate some more personas for the challenge and demonstrate how your tool can solve problems and add value to your users.

Persona 1: Fatima, 34 — Working Mother in Downtown Dubai

- **Profile:** Mid-level marketing executive, lives with her husband and two kids.
- **Needs:** Wants to save consistently for a future mortgage, but has fluctuating childcare and school fee expenses.
- **Challenge:** Hard to manage savings while still enjoying family outings and self-care.

Persona 2: Omar, 28 — Freelancer in Dubai Marina

- **Profile:** Freelance graphic designer with unpredictable income.
- **Needs:** Wants to buy a property in the next 3 years.
- **Challenge:** Finds it hard to decide when to save or spend, especially with varying project-based earnings.

Persona 3: Reem, 45 — Executive in DIFC

- **Profile:** High-income earner, single, but has fluctuating luxury lifestyle spending.
- **Needs:** Wants a smart savings tool that helps balance indulgence with long-term planning.
- **Challenge:** Her spending habits change seasonally due to travel, events, and fashion trends.

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LOGISTICS

LOGISTICS - HACKER BRIEF

MASTER QUESTION	TRADITIONAL HEALTH LOGISTICS DO NOT ADJUST TO DEMAND IN PREDICTIVE AND REAL TIME WAYS. How can agentic ai logistics assistant integrate hospital, gp and social media data to predict and respond to outbreaks and fluctuations in demand?
WHAT ARE WE OPTIMISING THE SYSTEM FOR?	A future where healthcare supply meets real-time demand! Prevent medicine shortages in high-risk areas. Reduce waste by optimising stock redistribution. Support faster response to outbreaks and seasonal health trends. Improve efficiency and cost-effectiveness in healthcare logistics.
WHAT ARE THE KEY USE CASES?	"I am a logistics manager for a national pharmacy chain. I need a system that anticipates demand for pharmaceuticals based on real-time health trends."
SOURCE DATA	DX Open Data, Synthetic Data
ACCESS	TBD
PROCESSING	Agentic AI-powered health logistics assistant that: Integrates hospital, GP, and social media data to predict outbreaks and medicine demand. Automatically adjusts supply chains to ensure the right pharmaceuticals reach the right locations. Minimises waste by redistributing surplus medicines to high-need areas. Provides insights and reports to stakeholders on health trends and inventory management.
TOUCH POINTS / INTERFACES	Web, mobile, smart wearables.
STARTERS FOR TEN	Real-time Health Monitoring: AI analyzes illness trends from multiple data sources, clinical and syndromic. Dynamic Logistics Optimization: Adjusts pharmaceutical distribution automatically. Proactive Public Health Support: Ensures medicine availability before shortages occur.
STRETCH GOAL	Add suppliers data
CONSIDERATIONS	Automatically adjusts orders back up the chain

LOGISTICS - DATA SOURCES

We encourage teams to spend at least 1 day compiling data their Agentic AI systems will need. Whether this be pulling from open datasets or creating synthetic data to work on all routes are accepted and encouraged so long as they are in the public domain. Here is some specific guidance for each project:

For the logistics challenge using real healthcare and logistics data is not possible so teams must synthesize their own data or find open datasets. We have compiled some datasets.

Datasets

Example kaggle dataset of healthcare data: [link](#)

Guidance

Paper on generating healthcare data: [link](#)

GitHub project for generating synthetic patient data: [link](#)

Live Data Sources

Here is a list of potential live data sources teams can use. This list is by no means exhaustive and if access to any is restricted teams are encouraged to synthesize some data and create a mock API. Please don't put API keys in your source code on GitHub.

Source Type	Details
World health info APIs	WHO API
Drug database API	FDB drug database
Mapping API	Google maps , OpenStreetMap
Live news API	Bing news API

LOGISTICS - PERSONAS

We have created 3 personas per challenge to give you some inspiration and guidance as to the types of challenges you will be solving. We would like teams to generate some more personas for the challenge and demonstrate how your tool can solve problems and add value to your users.

Persona 1: Dr. Ayesha, 42 — Public Health Director, Dubai Health Authority

- **Profile:** Oversees health policy and emergency response.
- **Needs:** Wants real-time data insights to prevent medicine shortages and respond to health spikes.
- **Challenge:** Current supply chains can't react fast enough to data from GPs, ERs, and public reports.

Persona 2: Imran, 39 — Regional Logistics Manager, Pharmaceutical Distributor

- **Profile:** Manages medicine delivery and stock levels across UAE.
- **Needs:** Wants predictive analytics to optimize delivery routes and stock priorities.
- **Challenge:** Misses emerging demand due to outdated reporting cycles.

Persona 3: Leila, 31 — Pharmacist in Deira

- **Profile:** Works at a busy community pharmacy.
- **Needs:** Wants to avoid running out of common medicines, especially during sudden surges in flu or allergy seasons.
- **Challenge:** No visibility into upcoming public health trends.

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ESTATE AGENT

INDUSTRY CHALLENGE



ESTATE AGENT - HACKER BRIEF



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MASTER QUESTION	How can Agentic AI continuously monitor rental markets, identify engaging opportunities or risks and proactively recommend investment and pricing strategies?
WHAT ARE WE OPTIMISING THE SYSTEM FOR?	Help decision makers get educated insights on real estate trends and have conversational optimisation of insight with Agentic AI assistants, to get the best special value of properties
WHAT ARE THE KEY USE CASES?	Large Real Estate Companies need to calculate property value, before taking decisions on where to invest and how to value their existing property rent
SOURCE DATA	e.g. property finder (public) and property monitor (private login required)
ACCESS	Anything open source
PROCESSING	Real Estate Market Price Screening Tool Crawl property market sites and listings e.g. property finder (public) and property monitor (private login required) to answer queries related to property market e.g. what are average property rental prices for 3 bedrooms in downtown Dubai, what are average property rental prices for 3 bedrooms in x,y,z neighbourhoods, what are average sizes of 2 bedroom apartments, which neighbourhoods have average 1 bedroom apartment sizes between xx sqm and yy sqm (we can develop further standard queries if required based on most typical data required)
TOUCH POINTS / INTERFACES	Web, mobile
STARTERS FOR TEN	TrendSpotter: scans listings for emerging rental shifts and growth hotspots ROI Forecaster: analyzes current and historical rents to predict yield opportunities PriceAdvisor: chats to recommend optimal rents and value-add strategies
STRETCH GOAL	Suggest multiple variants and value increase strategies
CONSIDERATIONS	Tenant Agents giving same

ESTATE AGENT- DATA SOURCES



Teams are encouraged to spend at least 1 day compiling good datasets and getting access to relevant APIs. If datasets or APIs aren't available teams are encouraged to explore web scraping as a means to acquire relevant data.

Datasets

Dubai tenancy agreement dataset: [link](#)

Dubai real estate portal: [link](#)

Explore [kaggle](#) for many open source datasets

Live Data Sources

There are not many APIs available for the real estate market so teams are encouraged to explore web scraping as a means to acquire the relevant live data. Here are some APIs we found:

DIFC real estate API: [link to press release](#)

Openstreetmap API: [link](#)

Mashvisor API: [link](#)

Zillow real estate metrics API: [link](#)

ESTATE AGENT- PERSONAS

We have created 3 personas per challenge to give you some inspiration and guidance as to the types of challenges you will be solving. We would like teams to generate some more personas for the challenge and demonstrate how your tool can solve problems and add value to your users.



Persona 1: Robert, 45 — International Portfolio Investor

- **Profile:** Dubai-based investor with a diversified global real-estate portfolio.
- **Needs:** Automated alerts on undervalued neighborhoods and shifting rental yields to guide acquisitions.
- **Challenge:** Manual, multi-market monitoring is too slow and fragmented to capture timely opportunities.

Persona 2: Amina, 29 — Residential Property Manager

- **Profile:** Oversees leasing and pricing for a 120-unit apartment building in Al Barsha.
- **Needs:** Dynamic rent-adjustment recommendations and proactive vacancy-management tactics.
- **Challenge:** Currently reacts only at lease renewals rather than adjusting rents in real time.

Persona 3: Faisal, 50 — CFO, Regional Real-Estate Developer

- **Profile:** CFO leading finance and acquisitions for a GCC mixed-use development firm.
- **Needs:** Regular scenario analyses and value-add investment proposals with projected ROI.
- **Challenge:** Reliance on static models and stale reports prevents agile strategy and forecasting.

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OWN CHALLENGE

OWN CHALLENGE - HACKER BRIEF

MASTER QUESTION	
WHAT ARE WE OPTIMISING THE SYSTEM FOR?	
WHAT ARE THE KEY USE CASES?	
SOURCE DATA	
ACCESS	
PROCESSING	
TOUCH POINTS / INTERFACES	
STARTERS FOR TEN	
STRETCH GOAL	
CONSIDERATIONS	